

Introduction of the Nervous system

Department Anatomy

Neuroanatomy (1)

Dr. osman Ahmed Osman

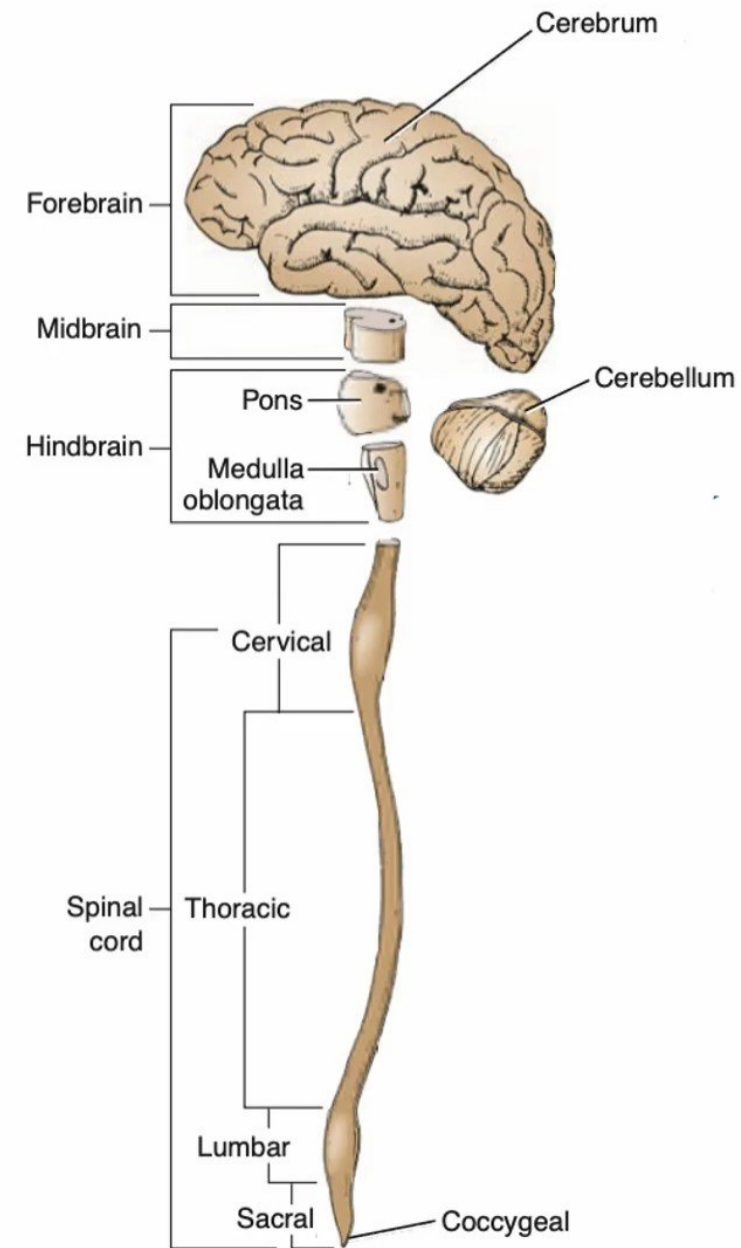
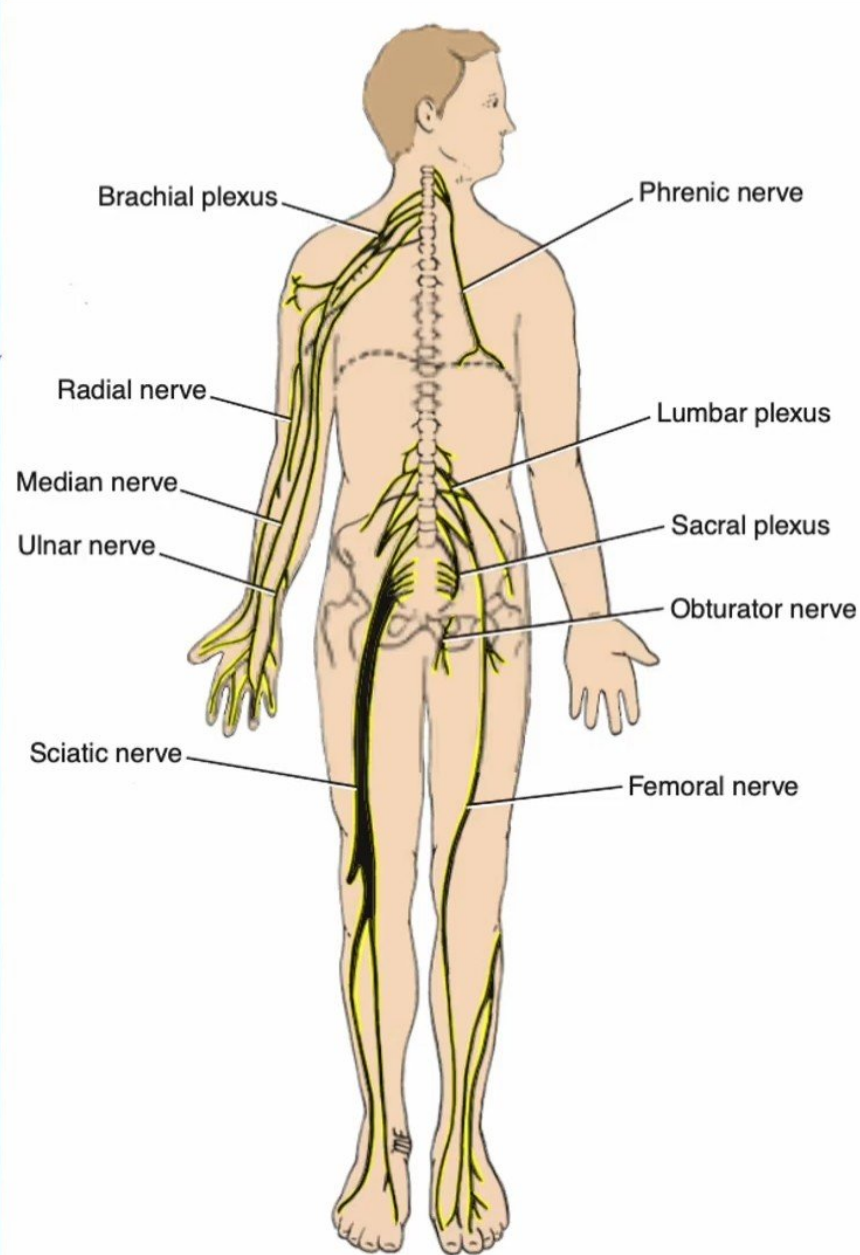
The nervous system

Anatomically can be Divided into:-

- 1) Central Nervous System
- 2) Peripheral Nervous System

Physiological can be Divided into:-

- 1) Somatic Nervous System
- 2) Autonomic Nervous System



Nervous System

```
graph TD;
    NS[Nervous System] --> CNS[Central Nervous System];
    NS --> PNS[Peripheral Nervous System];
    NS --> ANS[Autonomic Nervous System];
    CNS --> Brain[Brain];
    CNS --> SpinalCord[Spinal cord];
    PNS --> CranialNerves[Cranial Nerves];
    PNS --> SpinalNerves[Spinal Nerves];
    ANS --> Sympathatic[Sympathatic];
    ANS --> Parasympathatic[Parasympathatic];
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Central Nervous System

Brain

Spinal cord

Peripheral Nervous System

Cranial
Nerves

Spinal
Nerves

Autonomic Nervous System

Sympathatic

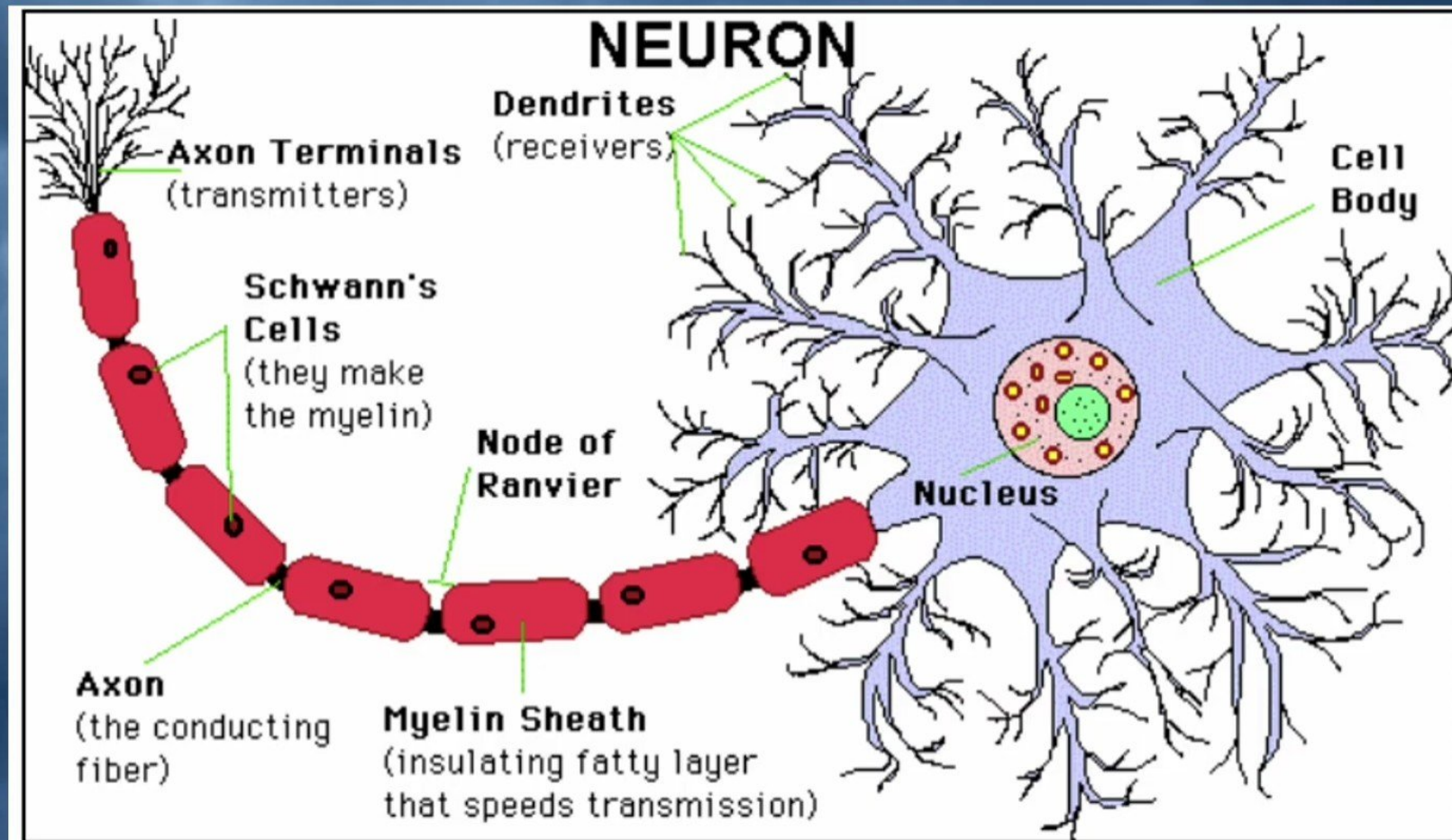
Parasympathatic

Histology of the nervous system

Cell types

1) Neuron:

- ✓ Conducts nerve impulses.
- ✓ Cannot be replaced if destroyed.

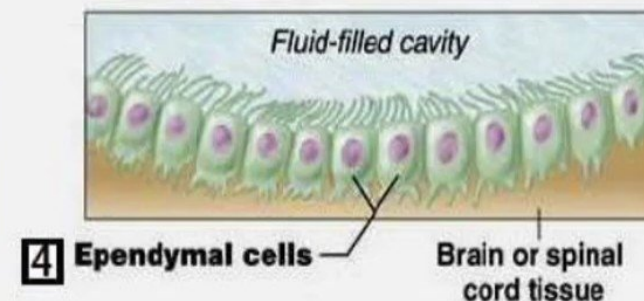
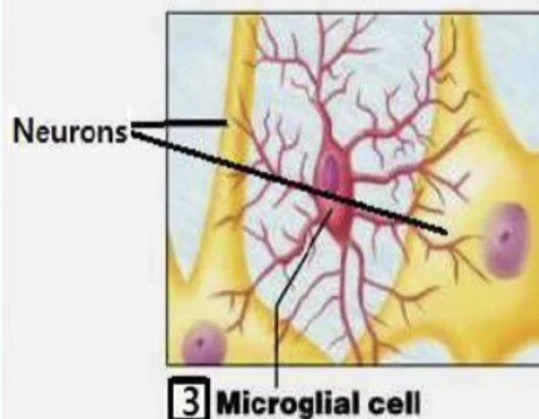
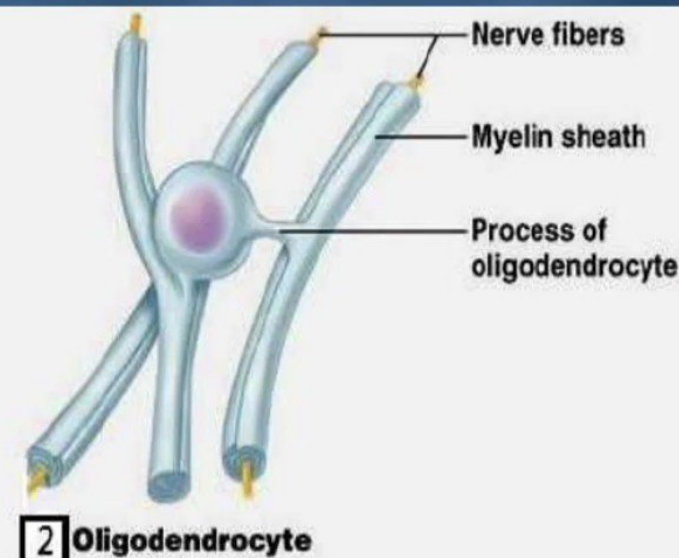
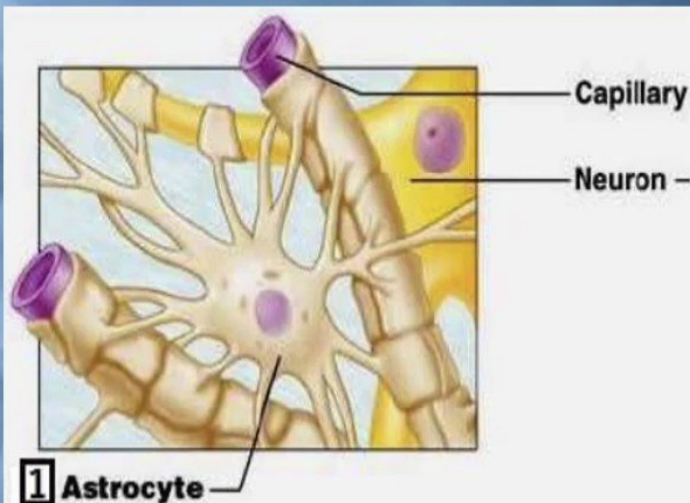


Histology of the nervous system

Cell types

2) Neuroglia:

- Supports, nourishes, and protects the neurons
- Includes **astrocytes**, **oligodendrocytes**, **microcytes** and **ependymal cells**.

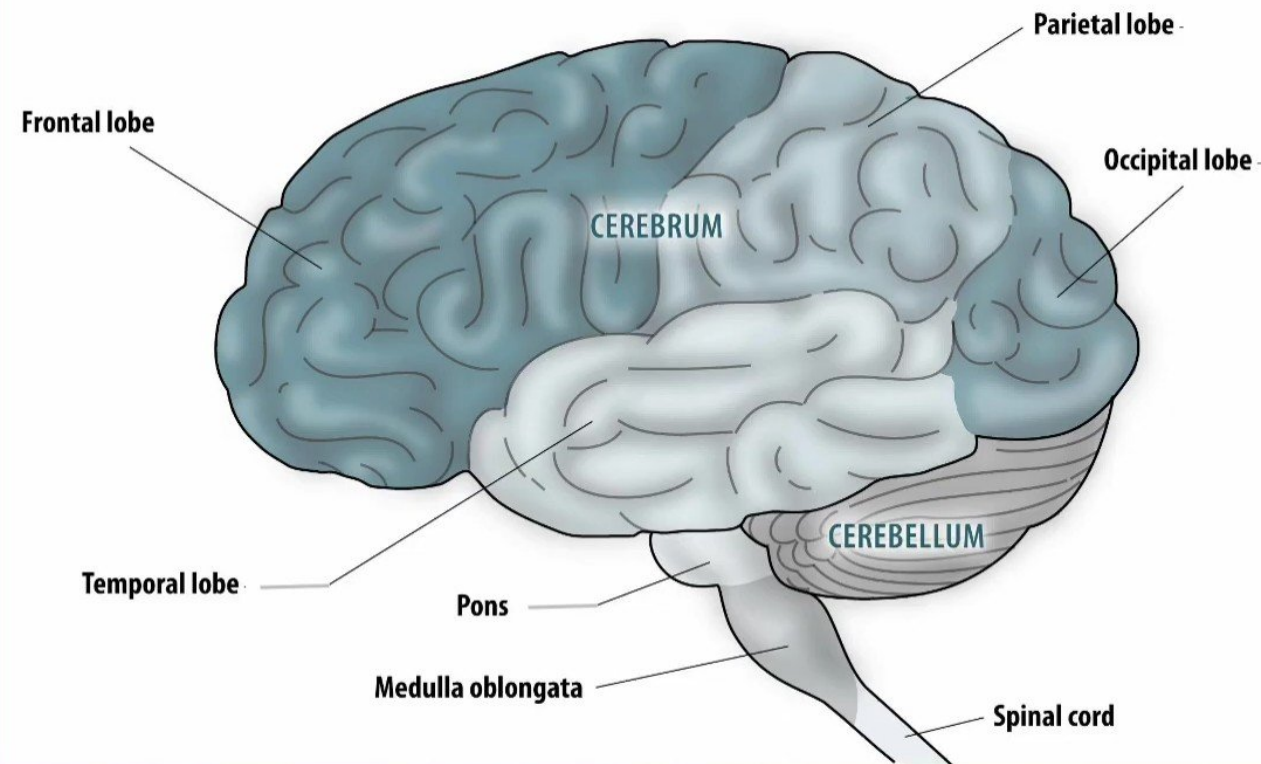


Cerebrum

- The cerebrum is the **largest** part of the brain and contains **two hemispheres**.
- The **right hemisphere** controls the left side of the body, and **the left hemisphere** controls the **right side of the body**.
- Each cerebral hemisphere contains **four lobes**; **the frontal lobe**, **the parietal lobe**, **the occipital lobe**, and **the temporal lobe**.

1) The frontal lobe controls **cognitive ability**, **memory**, **behavior**, **motor** and the ability to **speak** and **write**. Symptoms caused by a frontal lobe tumor include seizures, impaired judgment, personality changes, and short-term memory loss.

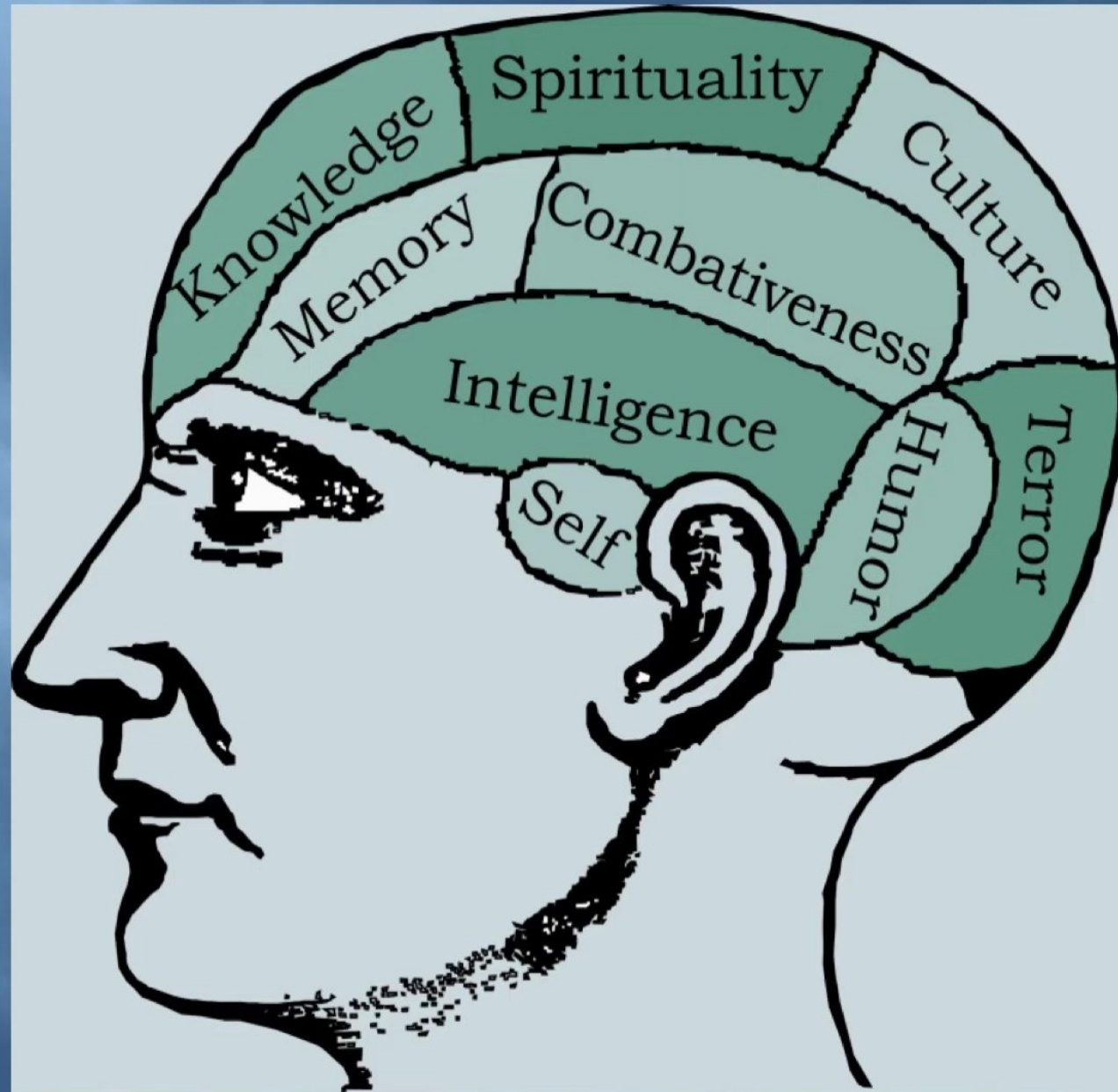
2) The parietal lobe controls **sensory discrimination** and **body orientation**. seizures, language disturbances, and the inability to do arithmetic are symptoms of a parietal lobe tumor.



3) The occipital lobe controls the **understanding of visual** images. Symptoms of a tumor in the occipital lobe include blindness in one direction and seizures.

4) The temporal lobe controls **hearing** and **the ability to understand the spoken word**, and the most common symptom of a tumor in the temporal lobe is **seizure**.

CNS Functional Anatomy



Parts of Brain Stem

The brain stem:

- It is at the bottom of the brain and connects the spinal cord to the cerebrum.
- The part of the brain stem are: pons, midbrain, and medulla oblongata.
- The brain stem controls blood pressure, heart beat, breathing, consciousness, and eating and sleeping patterns.

The cerebellum:

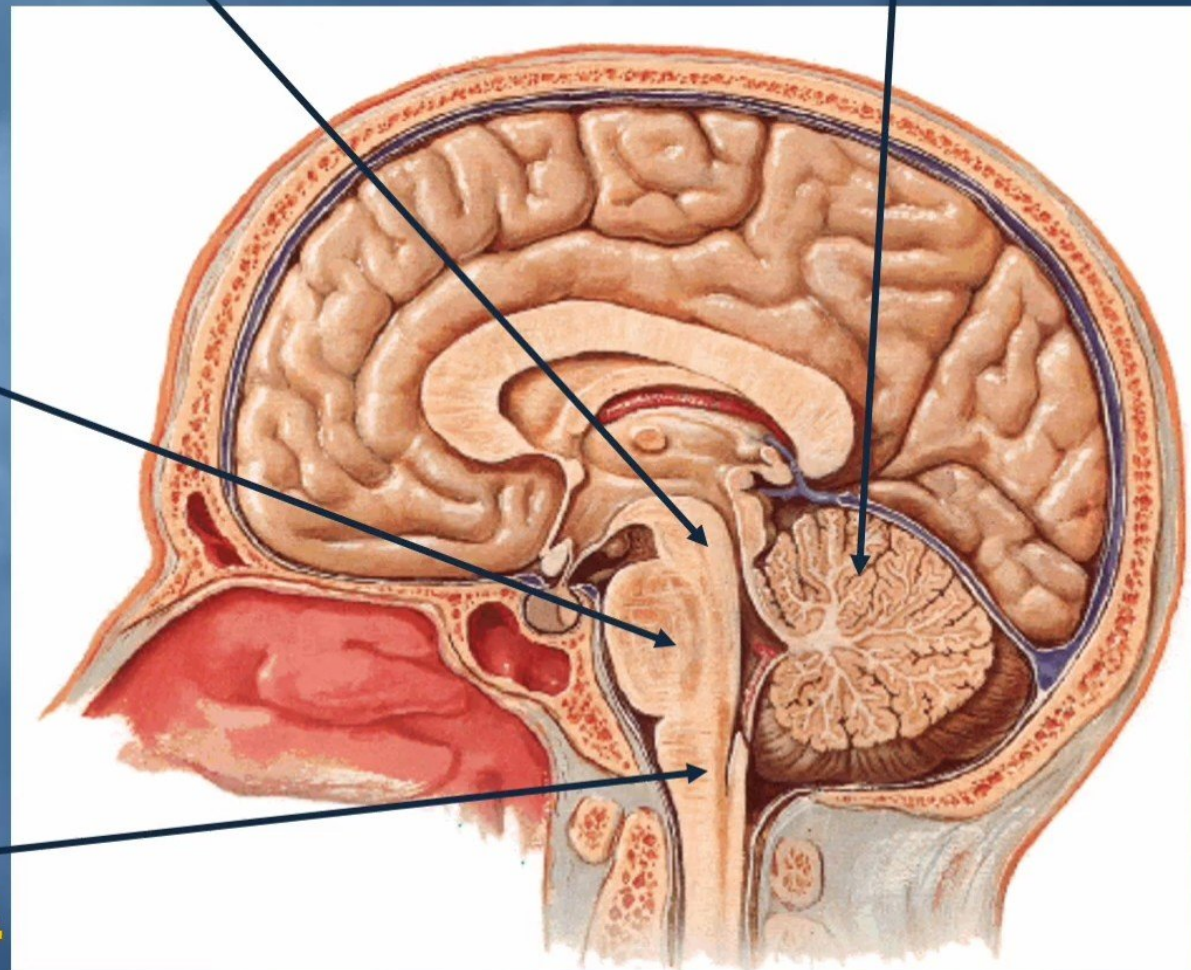
- It is the second largest area of the brain.
- It is involved with muscle coordination, walking, and speech.

Midbrain

The cerebellum

Pons

Medulla oblongata

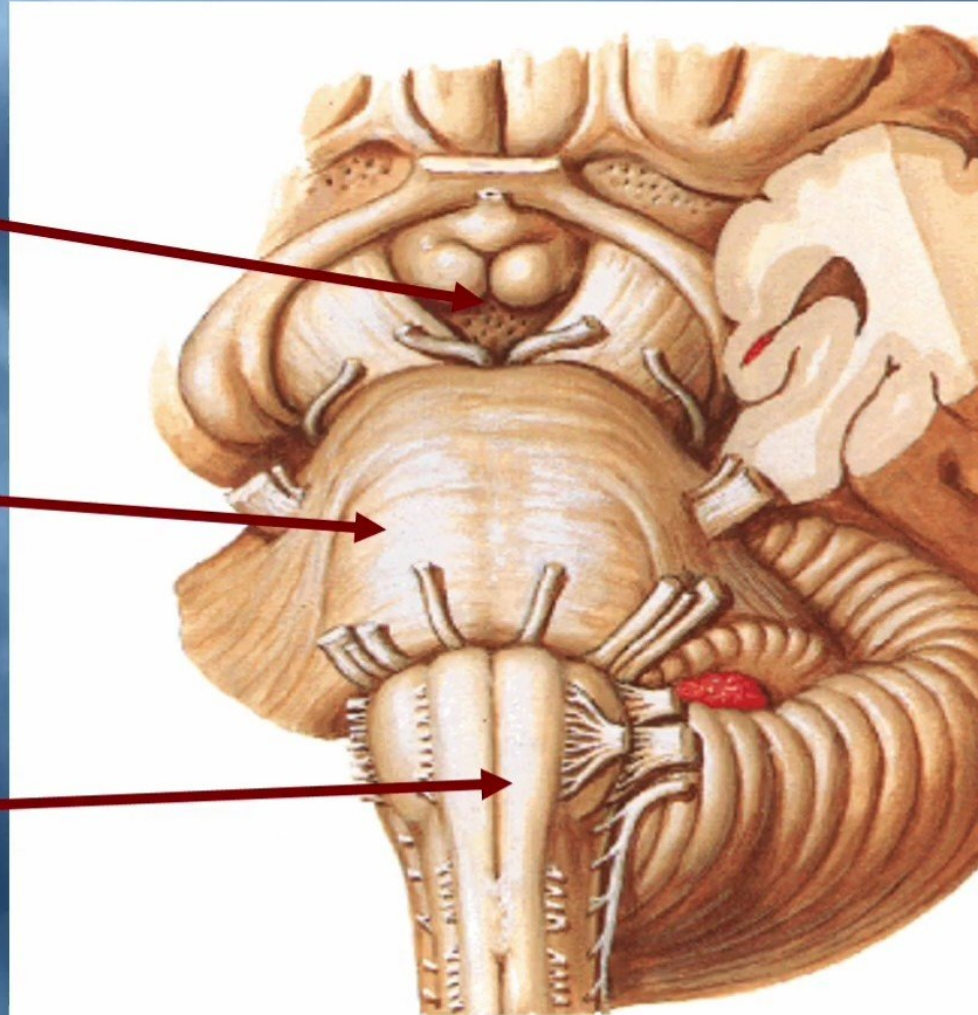


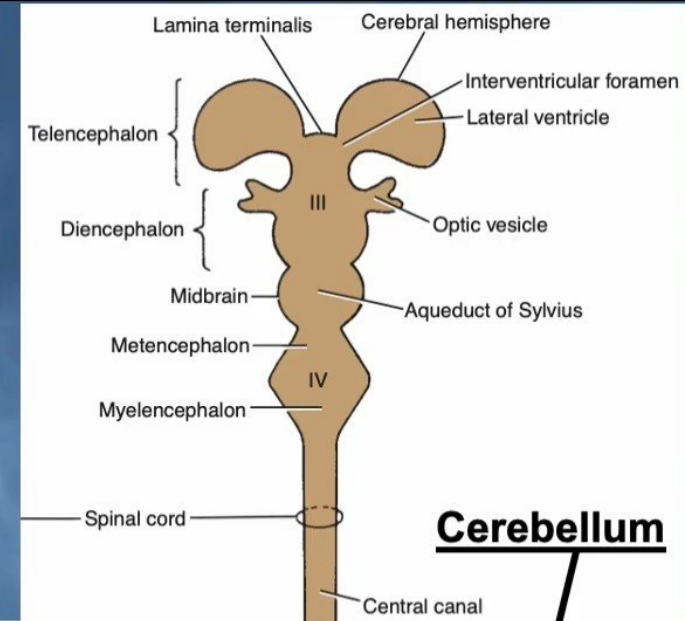
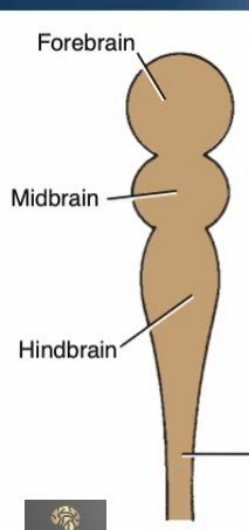
Brain Stem/Anterior View

Midbrain

Pons

Medulla oblongata





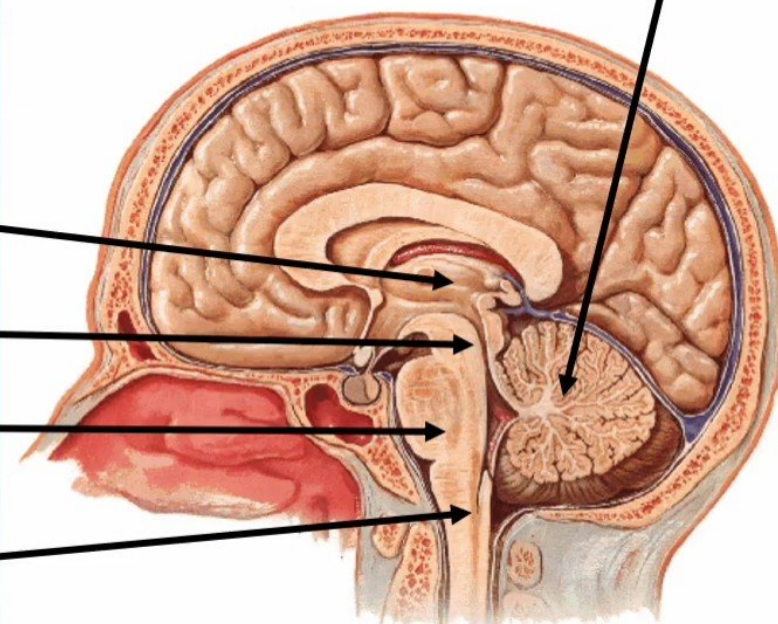
Diencephalon

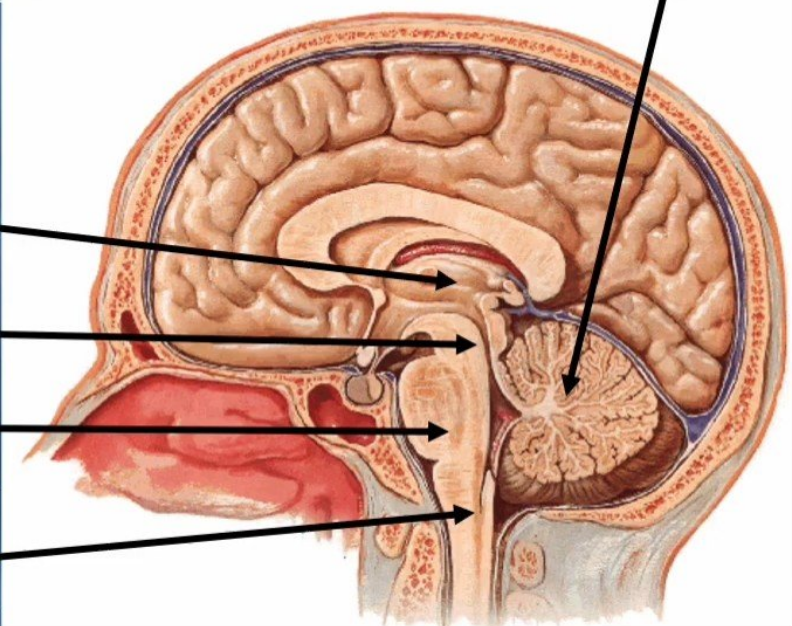
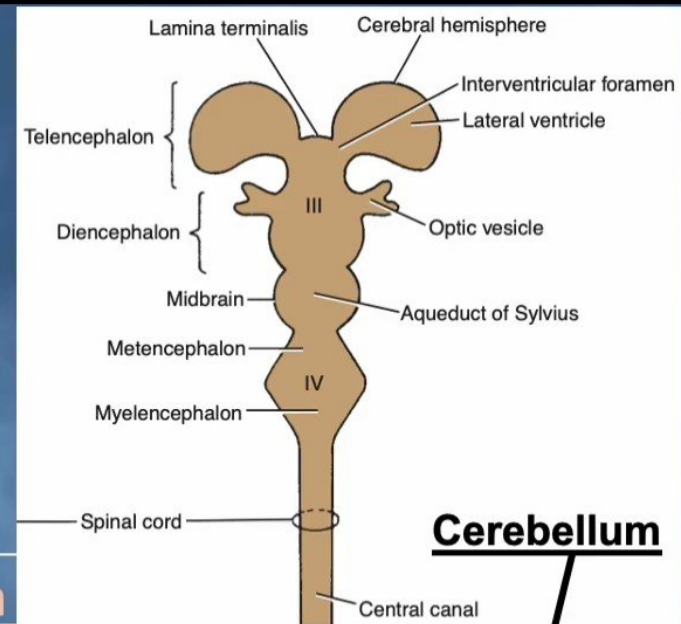
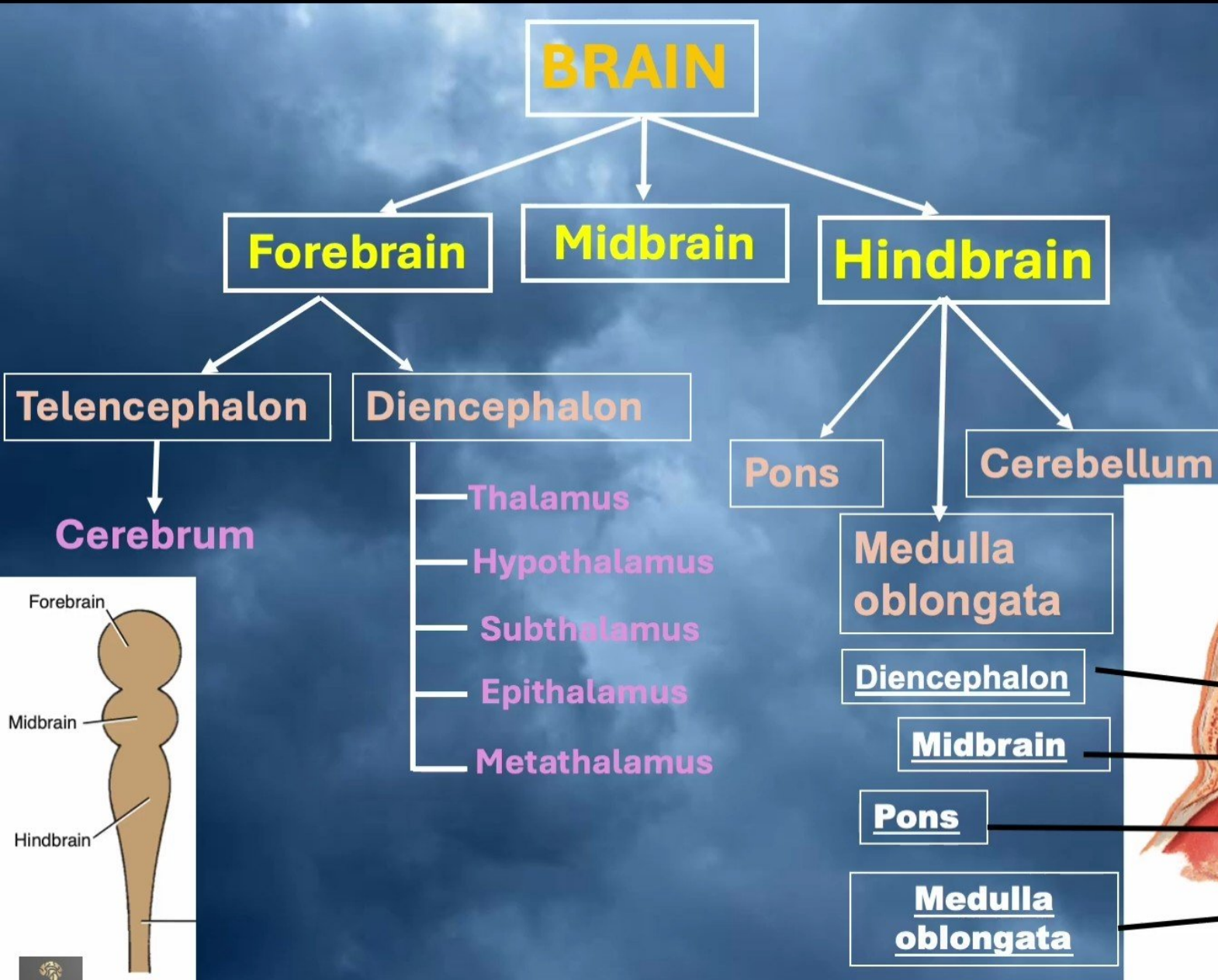
Midbrain

Pons

Medulla oblongata

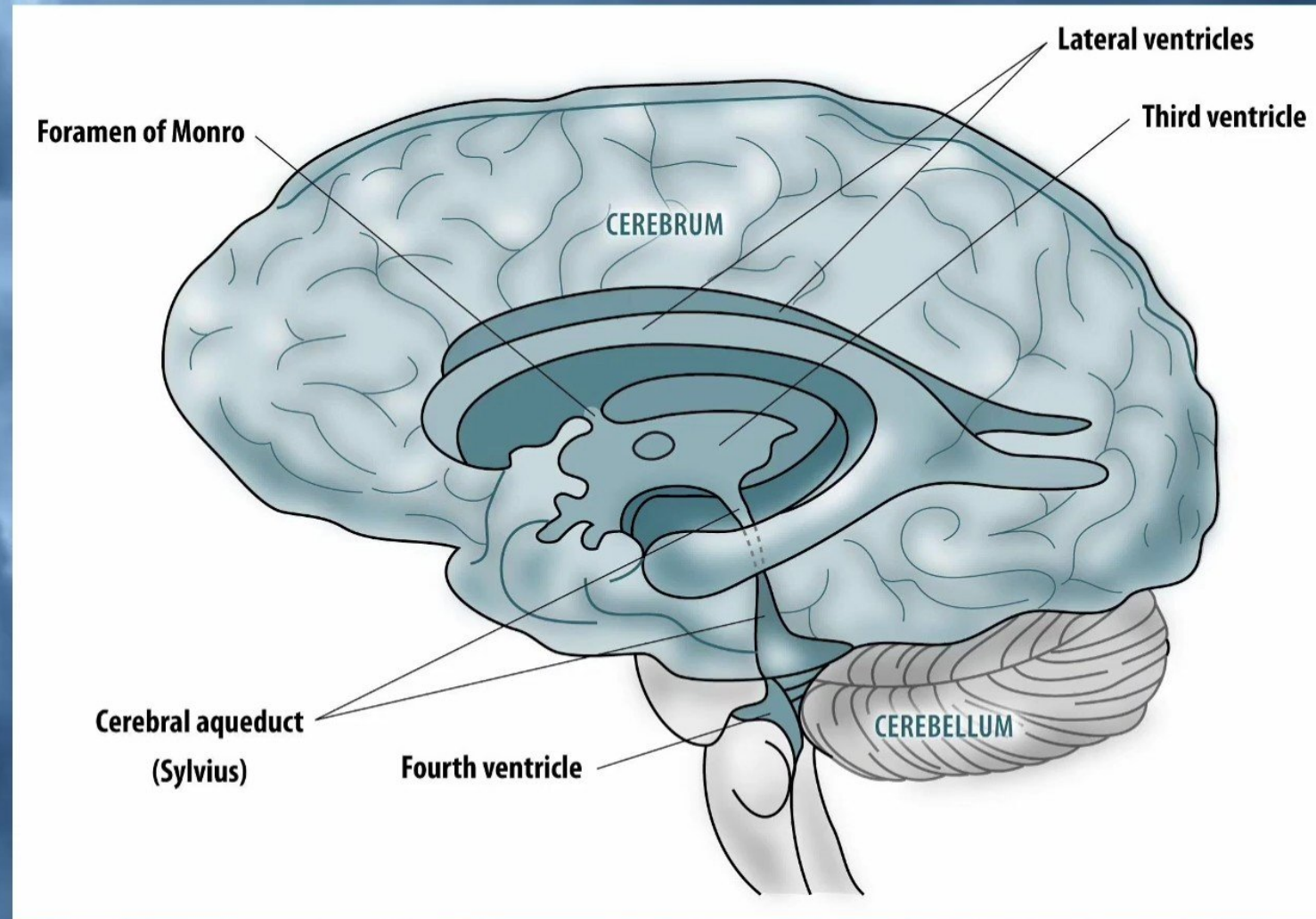
Cerebellum





Ventricular System

- The **ventricles** are **cavities within the brain** that contain **cerebrospinal fluid**.
- There are **two lateral ventricles**, one in each cerebral hemisphere.
- The **third ventricle** is located **near the hypothalamus and thalamus**, and
- The **fourth ventricle** is located between the **medulla oblongata** and the **brain stem** and the **cerebellum**.



Cerebrospinal Fluid

- Formed in the ventricles of brain.
- Weight of brain outside CSF = 1500gms.
- Weight of brain inside CSF = 50 gms.
- Stabilizing brain during head movement.
- Shock absorber.

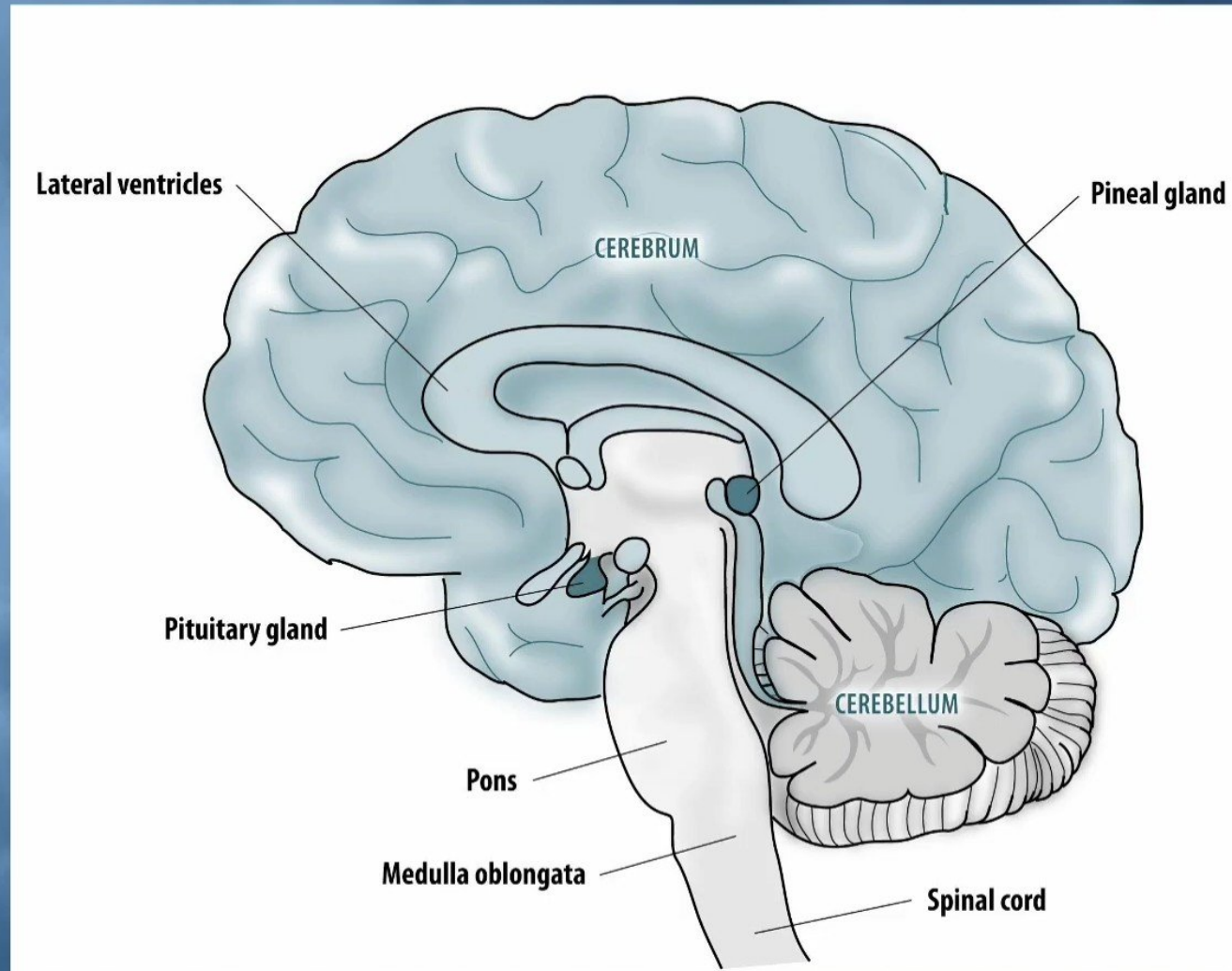
Pineal and Pituitary Glands

The pineal gland

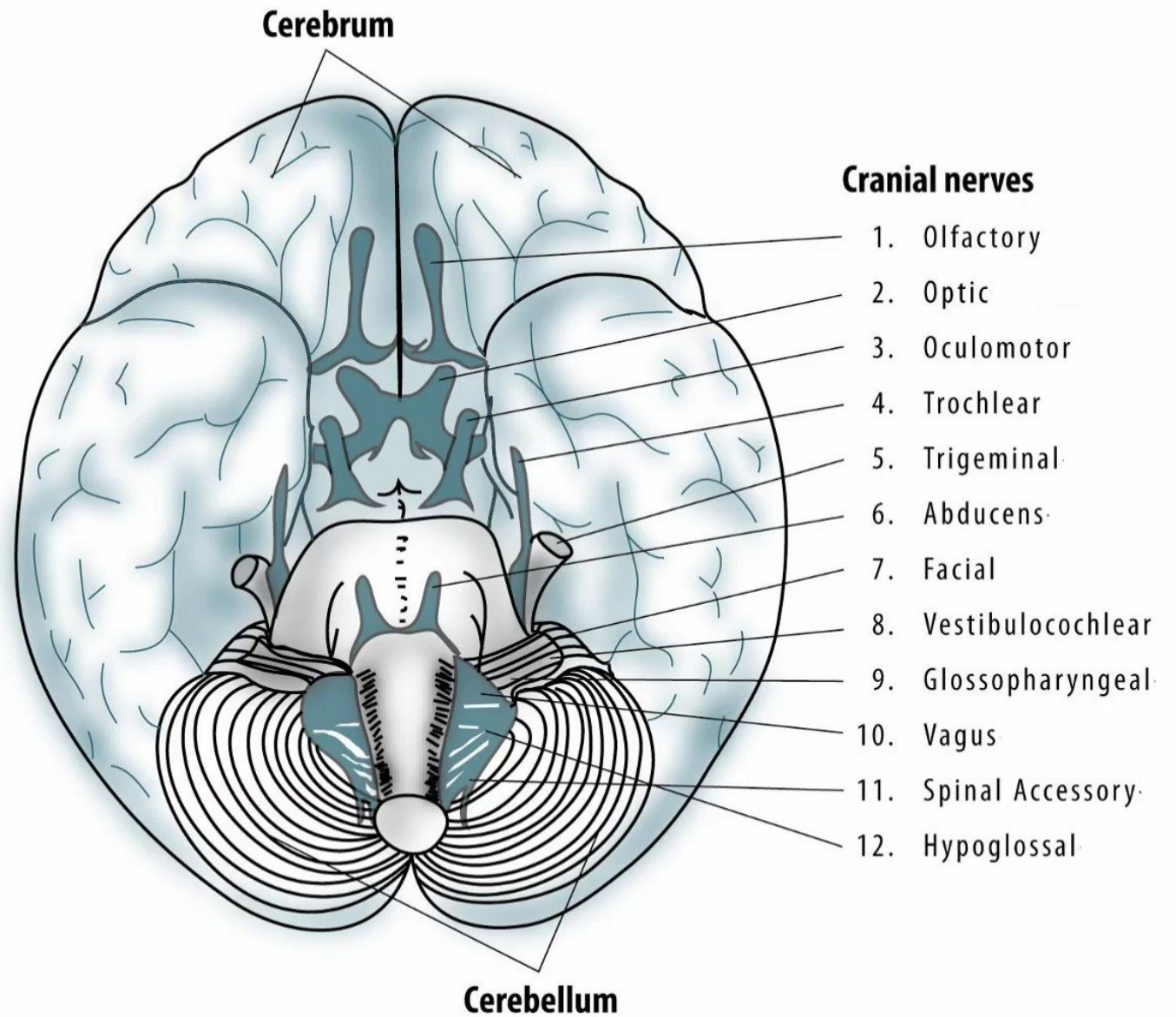
- It is located **deep inside** the brain.
- It **produces melatonin**, the hormone that controls **biological body rhythms**.
- **Hydrocephalus** is a symptom of a **pineal tumor**.

The pituitary gland

- It is located at **the base** of the brain.
- It **produces several hormones** including **growth hormone**.
- It **regulates and controls** other **hormone-secreting glands** (**thyroid** and **adrenal gland**).
- Common symptoms of pituitary tumors may include **diabetes insipidus**, **headache** and **vision** changes.



Cranial Nerves



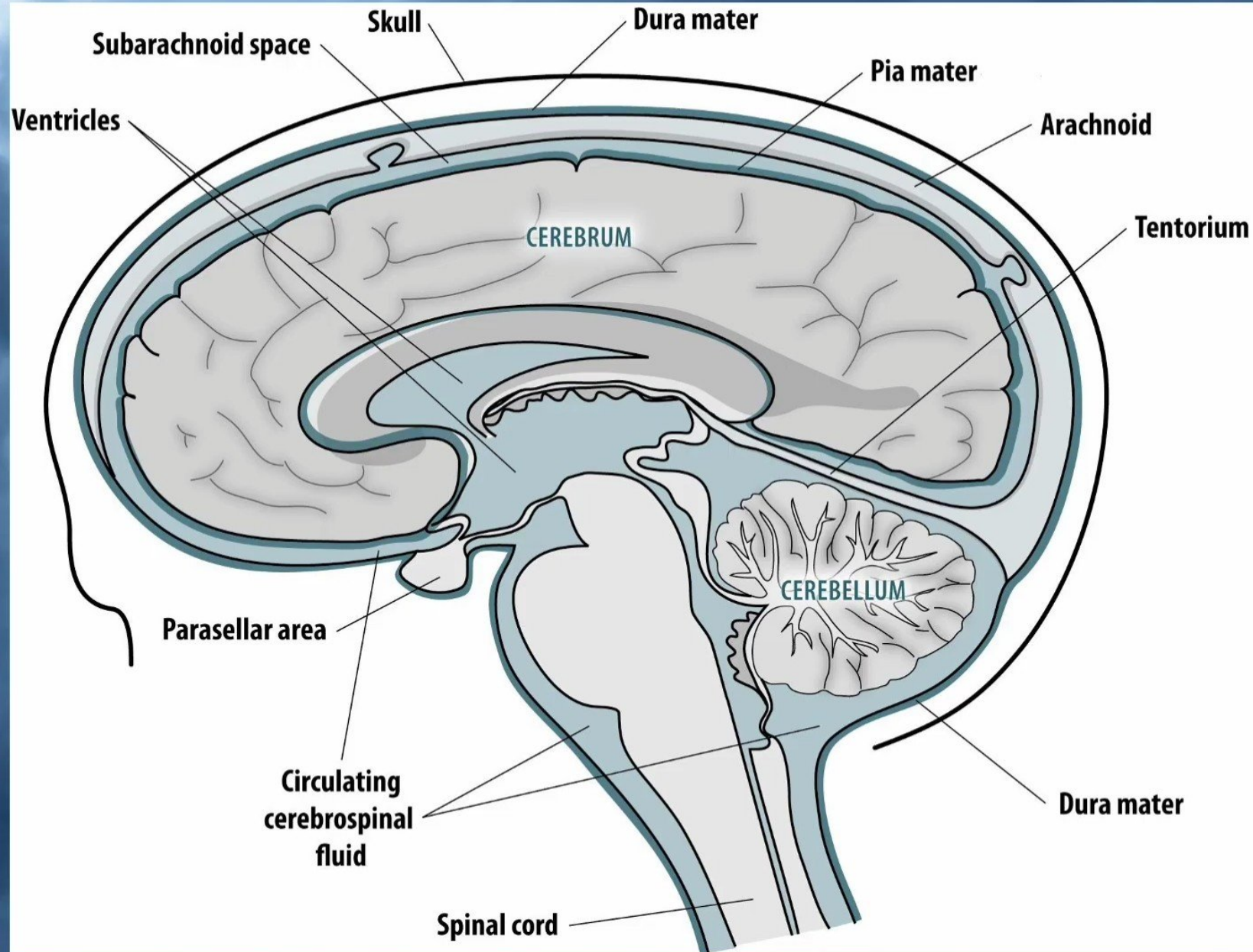
Cranial Nerves Functions

There are 12 pairs of cranial nerves:

- **The olfactory nerve (I)** controls sense of smell.
- **The optic nerve (II)** controls vision.
- **the oculomotor nerve (III)** controls eye movement and pupil size.
- **the trochlear nerve (IV)** controls eye movement.
- **the trigeminal nerve (V)** controls sensation in the face, nose, mouth, teeth, cornea, chewing, and facial expression.
- **The abducens nerve (VI)** controls eye muscles.
- **The facial nerve (VII)** controls facial expression, tears, and saliva taste.
- **The vestibulocochlear or acoustic nerve (VIII)** controls hearing and balance.
- **The glossopharyngeal nerve (IX)** controls throat movement and sensation and taste.
- **The vagus nerve (X)** controls sensation and muscles in the throat and windpipe.
- **The accessory nerve (XI)** controls movement of the neck; and the hypoglossal nerve (XII) controls tongue movement and swallowing.

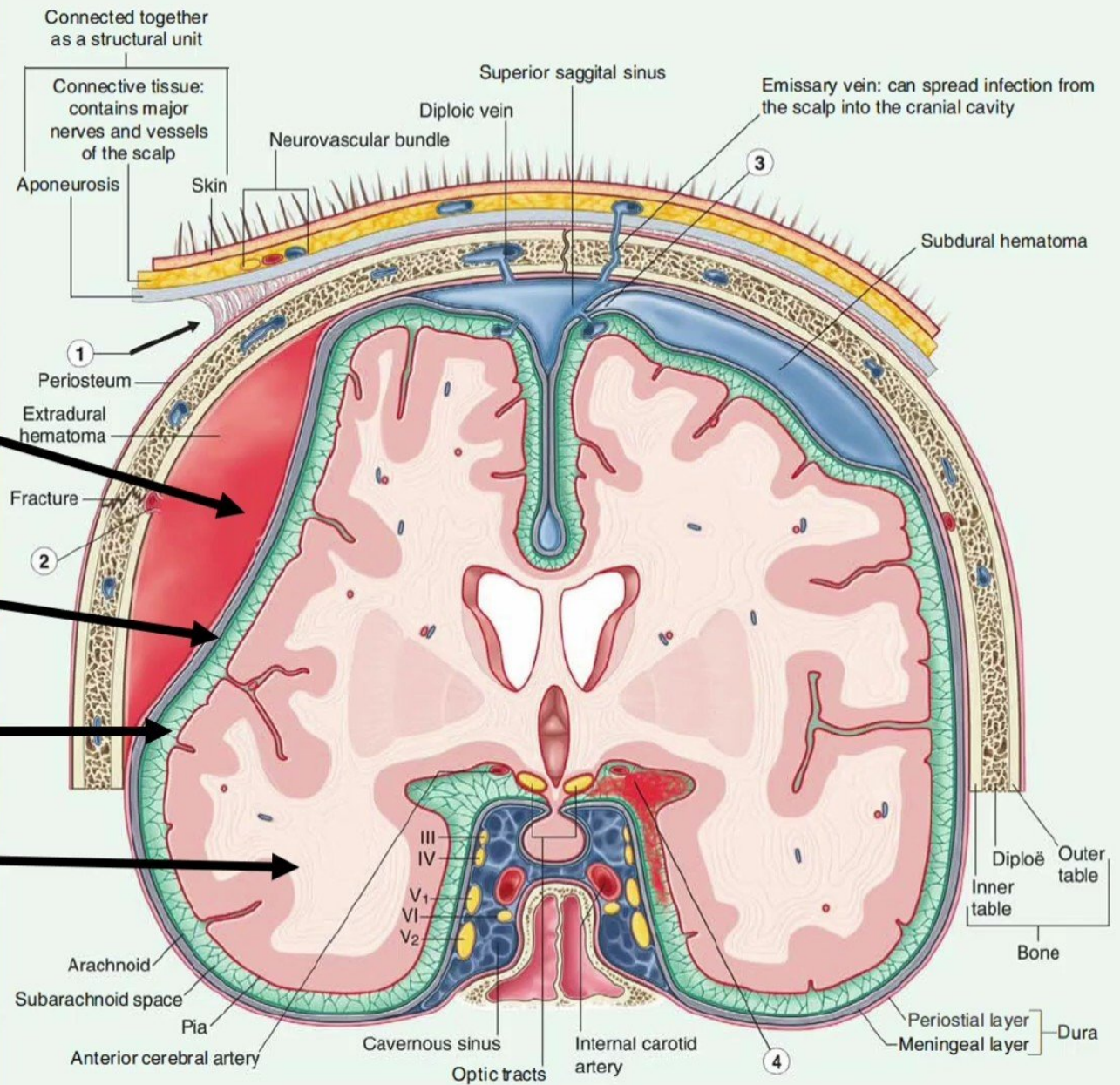
Meninges

- The meninges are three membranes that cover the brain and spinal cord and protect the CNS.
- **The dura mater** is the tough outer membrane.
- **The arachnoid** is the middle web-like membrane.
- **The pia mater** is the inner most membrane and is delicate and highly vascular.
- The subarachnoid space is between the arachnoid and pia mater and contains **cerebrospinal fluid (CSF)**.



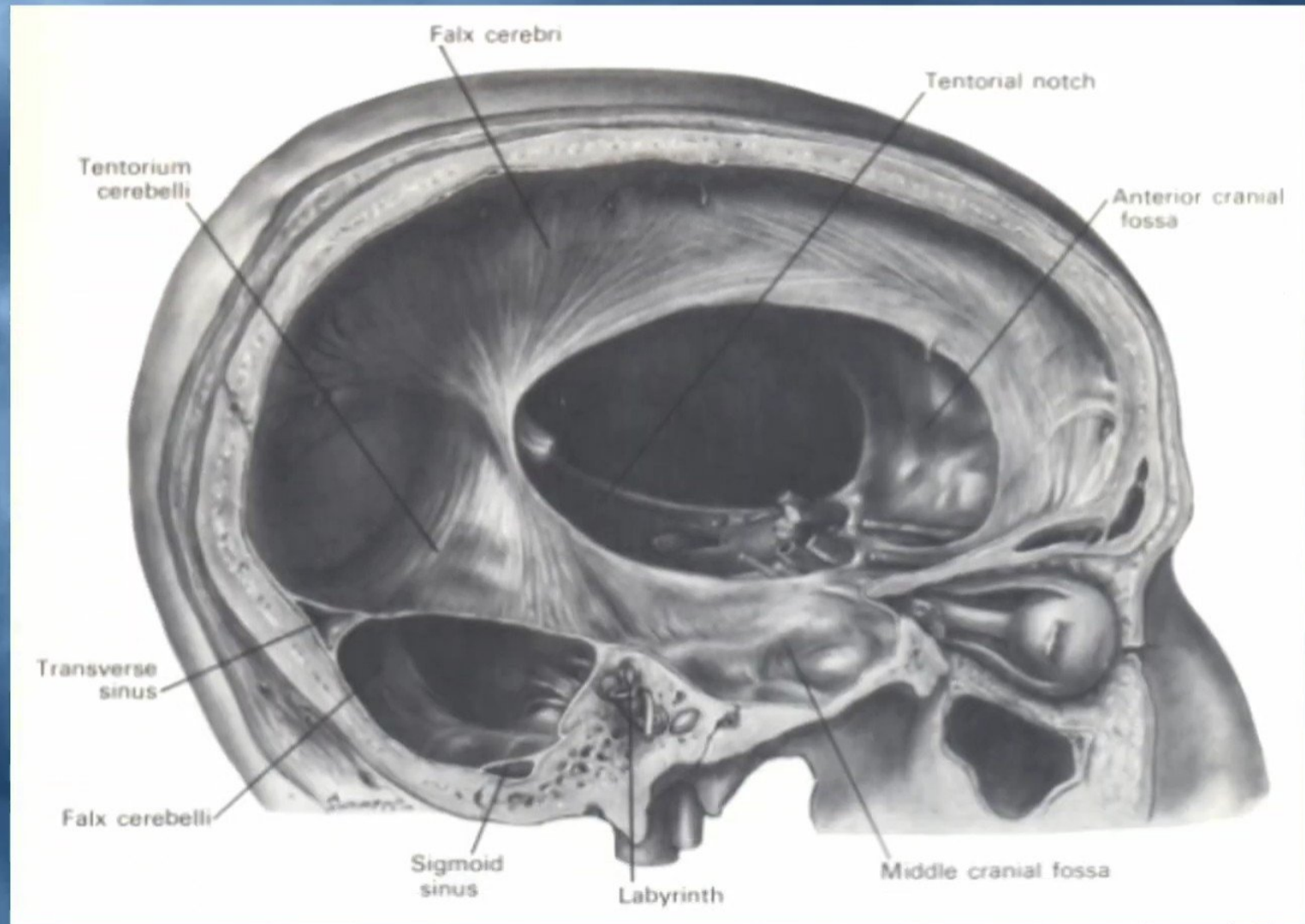
Meningeal Spaces

- Epidural space.
- Dura**
- Subdural space.
- Arachnoid**
- Subarachnoid space.
- Pia**
- Intracerebral.



Dural Reflections

- Falx cerebri
- Tentorium cerebelli
- Diaphragma sellae
- Falx cerebelli

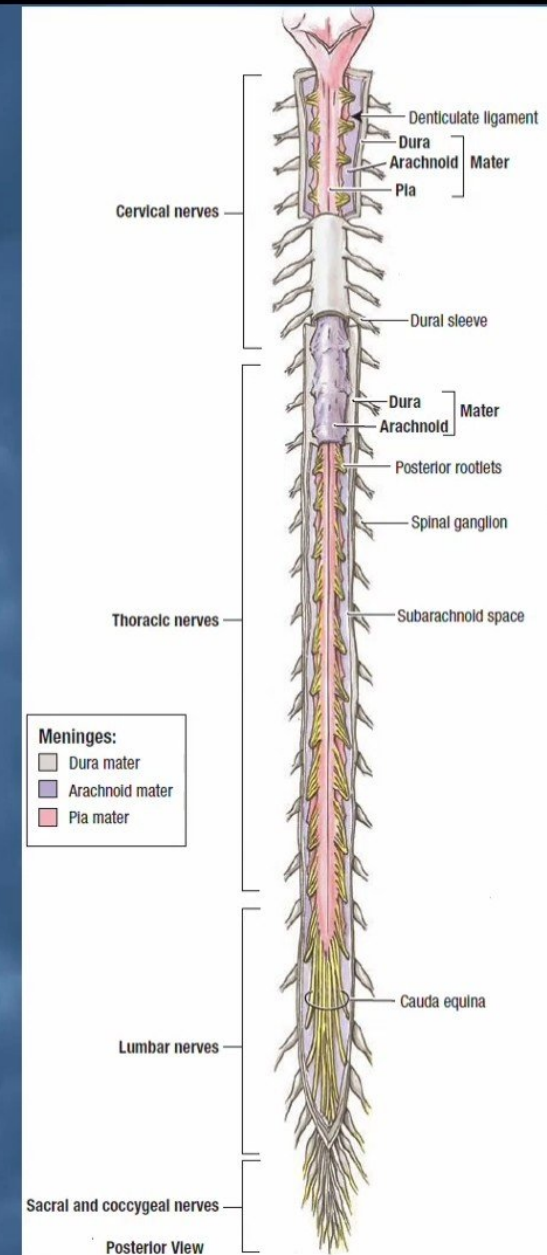


Spinal Cord

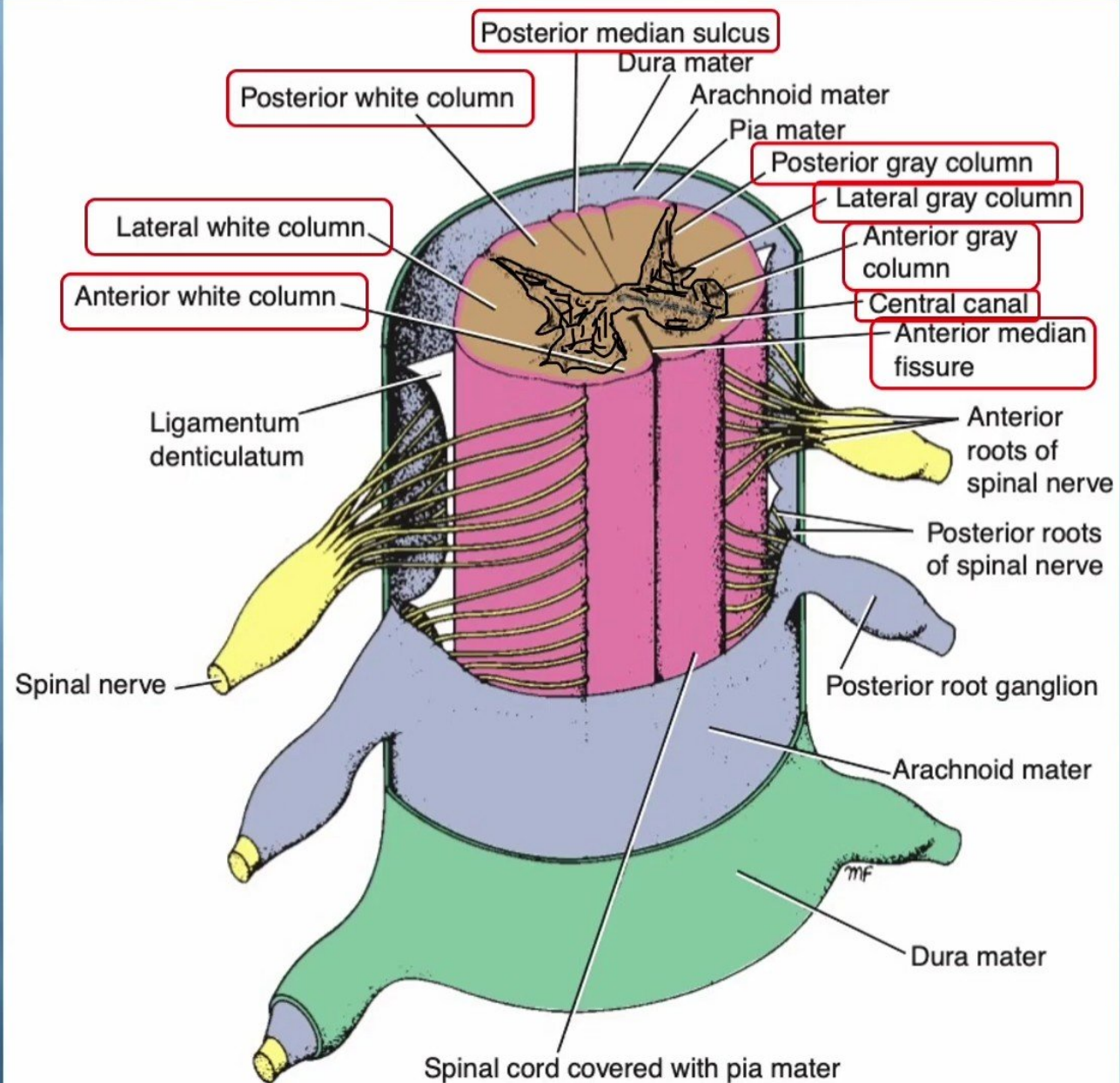
External features

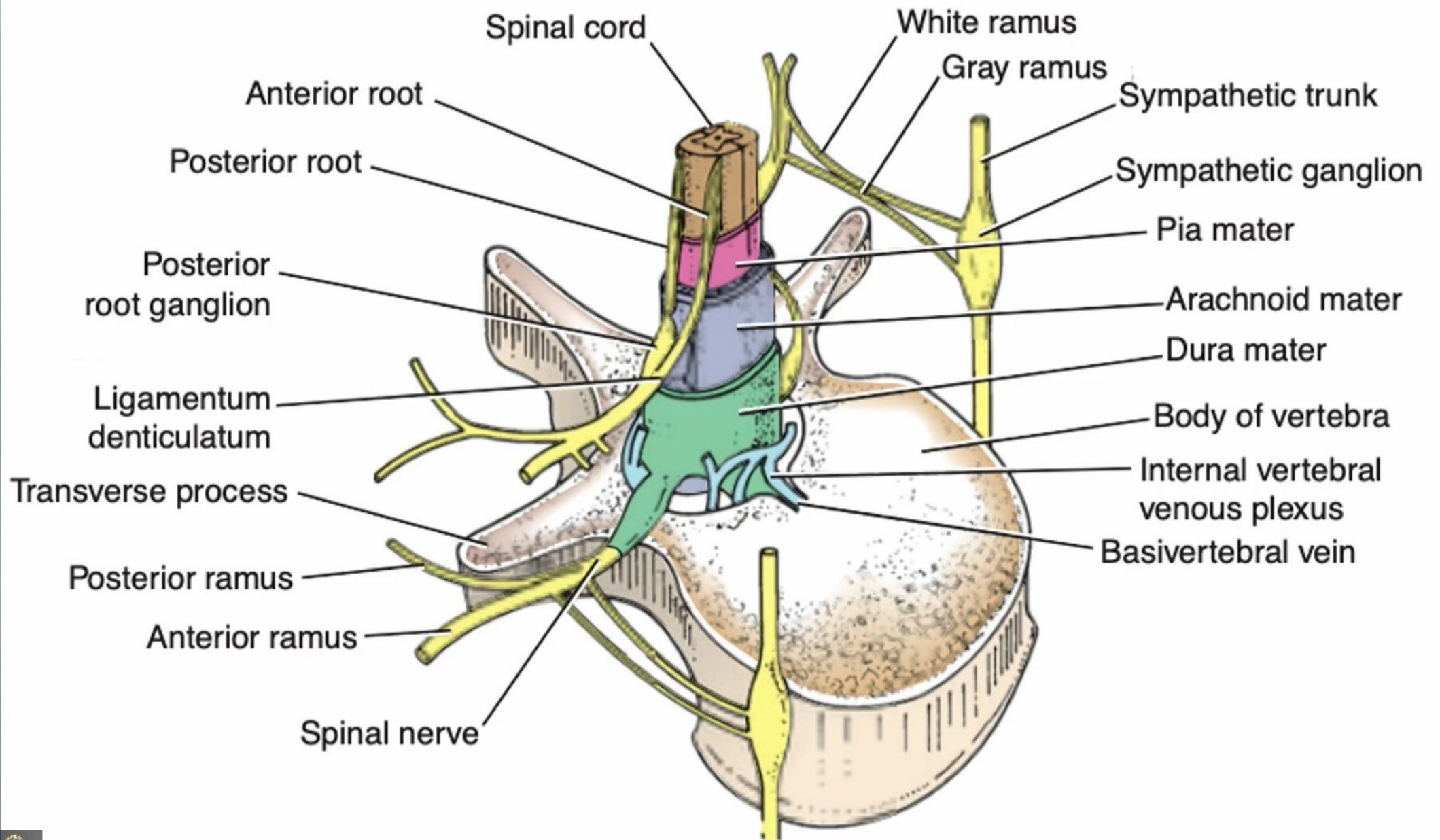
- Cylindrical in shape.
- Surrounded by three meninges :
 - ✓ (pia, arachnoid, dura mater).
- Length 45 cm.
- Weight 35 grams.
- Diameter is about 1 cm.

Internal features (Tractology)

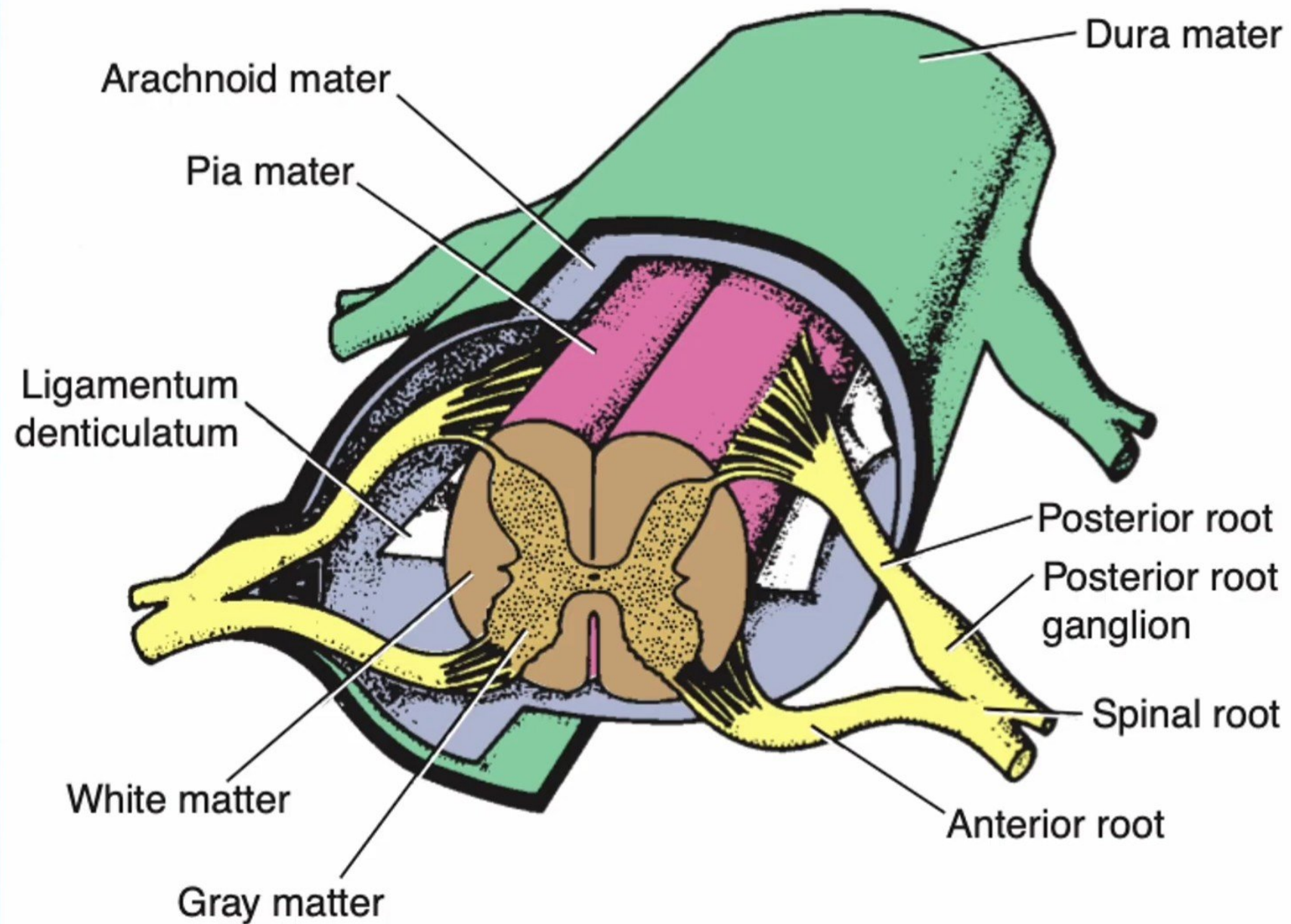


Internal Features

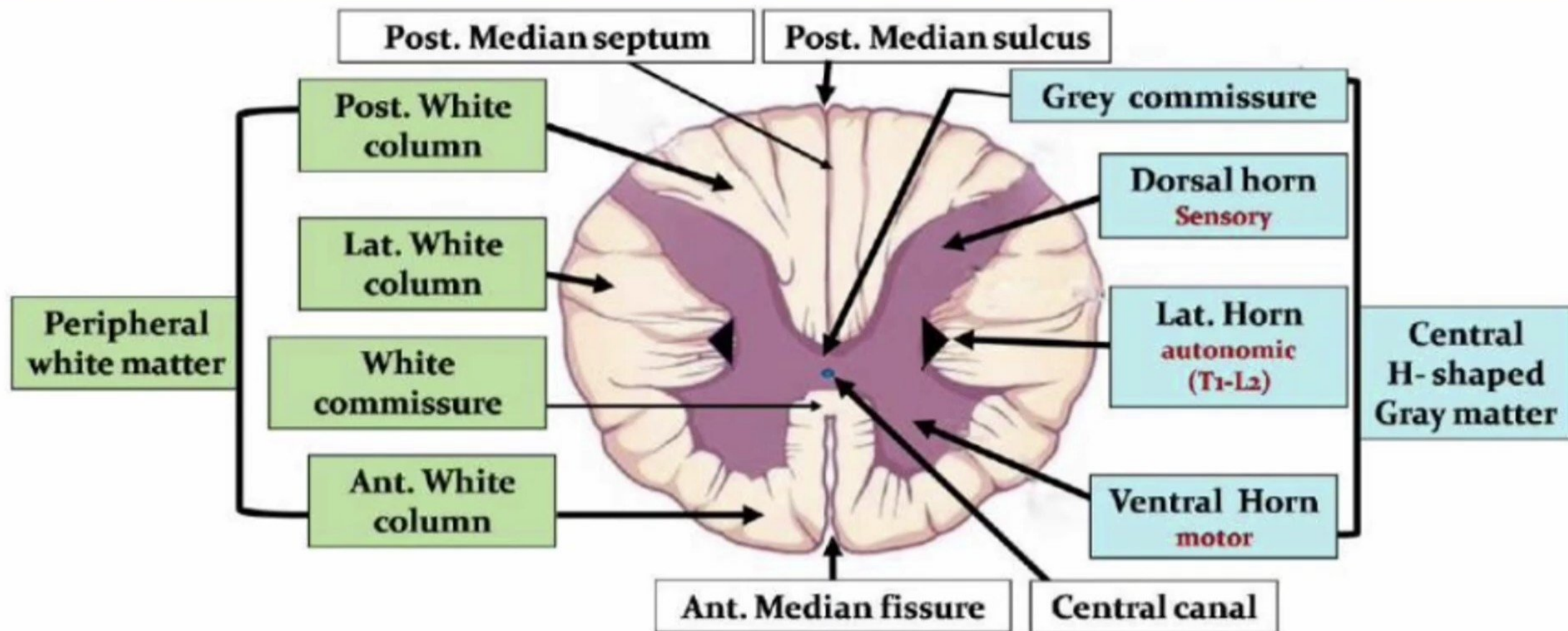




Internal Features and formation of spinal nerve



Internal Features (Tractology)



Thanks